Review retrospective:
the quartic–sextic numerator specialisation — one dead open, MeasureTheory stays

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Abstract

A soundness review of `Laplace/TwoD/QuarticSexticNumeratorAsymptotic.lean`: the $(k_1, k_2) = (2, 3)$ specialisation of the unnormalised 2D numerator $\int\int x^{2j} y^{2k} e^{-tL}$. Result: **a clean fidelity pass with one dead-open removal** — only `open Real`. Unlike the partition specialisation reviewed just before it, here `open MeasureTheory` stays load-bearing, because the numerator is stated as a bare $\int$.

1 Setting

The numerator counterpart of the partition specialisation, and the $(2, 3)$ instance of the generic numerator lift. It uses `integral_separable_addSeparable` and the 1D `quartic_moment_even`/`sextic_moment_even` to package the closed form, rescaled limit, and equivalence.

2 What was reviewed

Three public theorems plus the constant $C'(j, k) = \frac{1}{6}\, 24^{(2j+1)/4}\, 720^{(2k+1)/6}\, \Gamma(\frac{2j+1}{4})\, \Gamma(\frac{2k+1}{6})$, the $(2, 3)$ instance of the generic `kthKthNumeratorConst`, with exponent $-((2j + 1)/4 + (2k + 1)/6)$.

3 Fidelity / soundness findings

Sound; no theorem-level mismatch. Constant and exponent correct, the exact theorem carries `ht : 0 < t`, and the asymptotic theorems use eventual positivity at `atTop`. One report-only note, identical to the partition specialisation: the module docstring states the exact t-dependent identity without restating $t > 0$. Terse-but-true narrative, left as is — the recurring `TwoD/` module-prose pattern, best handled by a single batched hygiene tide.

4 Simplifications applied

One dead `open`. `open Real` is dead (all `Real.*` qualified, bare tˆ(real) is `rpow`). **And the contrast with the partition file:** here `open MeasureTheory` is *kept* — the numerator's statement contains a bare $\int z : \mathbb{R} \times \mathbb{R}, \ldots$, so the notation needs the open. The per-file check (the partition file used the `partitionFunction` def, this one a bare integral) is exactly what distinguishes "one dead open" here from "two" there. Module green at 2702 jobs, no warnings, signatures byte-identical. GPT affirmed.

5 Disagreements and open questions

None. The reviewer's `show`→`change` suggestions were declined as proof-golf.

6 Lessons

- **Two adjacent specialisations, two different dead-open counts.** The partition file (`partitionFunction` def, no bare $\int$) had `open MeasureTheory` dead; this numerator file (bare $\int$) keeps it. The dead-open verdict tracks *what notation the statement actually uses*, not the file's family — and a one-line grep for bare $\int$ settles it before the build does.

7 Follow-ups

- Remaining `QuarticSextic*`: `MomentAsymptotic` (Gibbs moment — likely bare expectation, `open MeasureTheory` possibly dead like the partition file), and the closed-form cores `QuarticSextic`/`QuarticSexticMoments`/`QuarticSexticOddMoments`.
- Batched hygiene tide: the `t>0` module-docstring clauses across `TwoD/`, plus the `show`→`change` proof-golf.